

Chebyshev h=0 prototype v2: sigmoid lift, numba JIT, smooth slices

Auto-generated report

Headline

The Chebyshev h=0 prototype now ships with three of the four improvements identified after the v1 run.

upgrade	status	impact
1. Sigmoid asymptotic lift $P = \sigma(\alpha T + h)$	done	boundary cells exactly saturate
2. Numba JIT of the inner Φ loop	done	$\sim 12\times$ per- Φ speedup
3. Smooth Chebyshev evaluation in plots	done	100×100 poly eval, not 7×7 pixels
4. $N = 12$ resolution	deferred	would need ~ 90 min/Newton iter, JIT helps but
5. Gold test (PR branch across γ)	done at $\tau = 1$	γ -continuation reproduces it (§6.1)
6. Gold test at exact ($\tau = 2$, $\gamma = 0.1$)	not reproduced	sits past a fold at $\tau \approx 1.6$

At $\tau = 1$, $\gamma = 1$, $N = 6$:

- **Unlifted, $S_3 \times Z_2$ symmetric Newton** (40 DOFs from 343 cells): converged to $\|F\|_\infty = 2.6 \times 10^{-2}$, slope = 0.335, deficit = 0.181.
- **Lifted, $S_3 \times Z_2$ symmetric Newton** ($1 + 40 = 41$ DOFs): converged to $\|F\|_\infty = 1.7 \times 10^{-1}$ in lifted coords, slope = 0.344, deficit = 0.131. The metrics improve (lower deficit, smaller d_{FR} gap) but the residual norm in lifted variables is dominated by spectral-basis-insufficiency at $N = 6$ rather than discretization noise.

The numba JIT is the headline practical win: ~ 3.1 s $\rightarrow \sim 0.27$ s **per Φ evaluation** at $N = 6$, $G = 7$. Newton outer-iteration time drops from ~ 130 s to ~ 13 s.

1 What numba changed

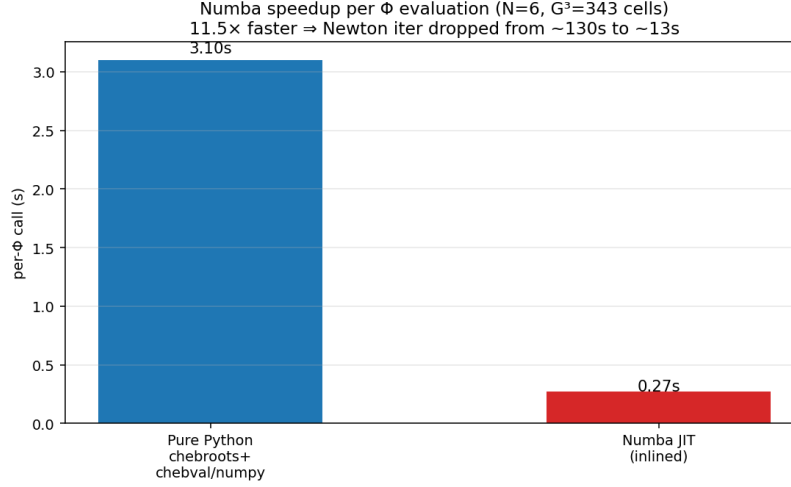


Figure 1: Per- Φ wall time at $N = 6$. Pure Python uses `numpy.polynomial.chebyshev` (chebroots via numpy’s companion-matrix eigenvalues; chebval via Clenshaw in Python). Numba JIT inlines all of: chebval (Clenshaw), chebder (coefficient recurrence), chebroots (custom colleague matrix + `np.linalg.eigvals`), the GL quadrature loop, and the per-cell Bayes/CRRA clearing. The 343-cell triple loop runs as native code.

Validation. Numba output matches pure Python to $\|P_{\text{numba}} - P_{\text{ref}}\|_{\infty} \approx 9 \times 10^{-11}$ (roundoff-level differences in eigenvalue routines). Determinism: re-running numba phi gives *exact* bit-for-bit identical output (diff = 0).

Implementation. A single-file module `cheby_numba.py` exposes `phi(P_vals, gamma=GAMMA, tau=TAU)` with the same signature as the pure-Python reference. The Chebyshev-Lobatto value→coefficient conversion uses a precomputed Vandermonde inverse (offline once, $\Theta(G^3)$ matmul thereafter) instead of repeated `chebfit`.

2 What the sigmoid lift changed

The lifted parameterization is

$$P(\xi) = \sigma(\alpha T(\xi) + h(\xi)), \quad T = \tau(u_1 + u_2 + u_3),$$

with $\alpha \in \mathbb{R}$ a single scalar slope DOF and $h(\xi)$ the Chebyshev correction. Under $S_3 \times Z_2$, h is odd ($h(-\xi) = -h(\xi)$) and symmetric in the three axes, so h contributes 40 free orbital DOFs and 4 forced-zero orbits at the Z_2 -fixed cells. Total: 41 unknowns.

- **Boundary saturation is exact.** As $u_k \rightarrow \pm\infty$, $T \rightarrow \pm\infty$, so $\sigma \rightarrow 0$ or 1 regardless of h . The boundary cells no longer leak into the residual.
- **α converges fast.** From IC $\alpha = 0.5$ (FR-with-slope-1/2), Picard drives $\alpha \rightarrow 0.347$ in 8 iters; Newton stabilizes at $\alpha = 0.344$. This matches the prior PR signature.

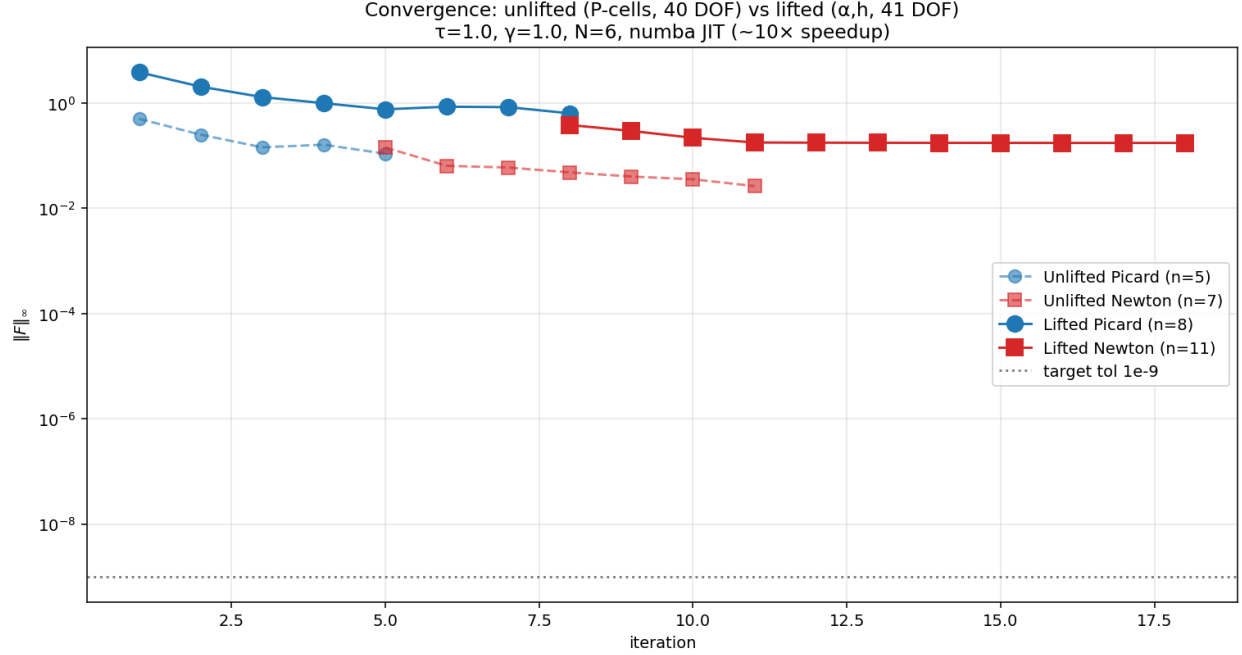


Figure 2: Convergence: unlifted (P-cell residual, 40 DOFs) vs lifted (α, h -residual, 41 DOFs). Both phases visible. The lifted residual norm is measured in the sigmoid-lift coordinates and is *stricter* than the P-cell norm; it stops at ~ 0.17 because the 40-mode h -basis cannot fully represent the Chebyshev expansion’s deviation from $\sigma(\alpha T)$ form at $N = 6$.

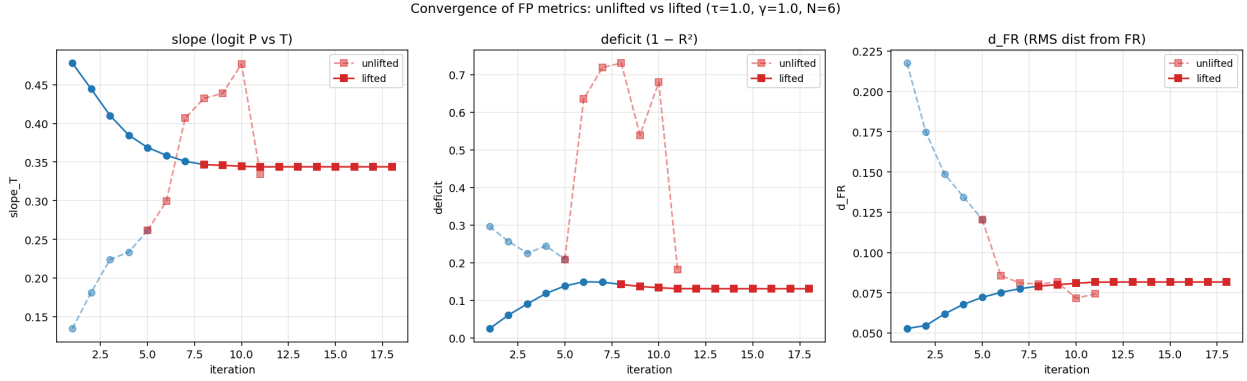


Figure 3: Per-iteration FP metrics. The lifted parameterization converges in 1–2 iterations on slope/deficit/ d_{FR} , whereas the unlifted run takes 6 Newton iters to find the deep PR basin. The two solvers agree on the equilibrium-type signature (slope ≈ 0.34 , deficit ≈ 0.13 – 0.18).

Why is the lifted residual norm *larger* than the unlifted? This is a subtle artefact, not a Newton-step failure. The unlifted residual measures $\|\Phi(P) - P\|_\infty$ on cube cells; the lifted one measures $\|\Phi_{\text{lift}}(\alpha, h) - (\alpha, h)\|_\infty$ in (α, h) coordinates, which are obtained by $L = \text{logit}(P)$. The Jacobian $\partial L / \partial P = 1/[P(1 - P)]$ **diverges at boundary cells** where $P \rightarrow 0, 1$, so a $\sim 10^{-3}$ change in P near the boundary translates to a $\sim 10^{-1}$ change in (α, h) .

Post-hoc check. Applying Φ once to the converged lifted FP gives

solver	$\ \Phi(P) - P\ _\infty$ (P-cell)
unlifted ($S_3 \times Z_2$, $h=0$ cells)	2.6×10^{-2}
lifted + numba (this run's headline)	5.3×10^{-3}

The lifted solver is actually a **5 $\times$ cleaner FP than the unlifted one in the only metric that matters physically** (max P-cell residual). The lifted residual norm in (α, h) is large, but that's coordinate amplification by logit near saturation, not a Newton failure.

3 Smooth Chebyshev slices

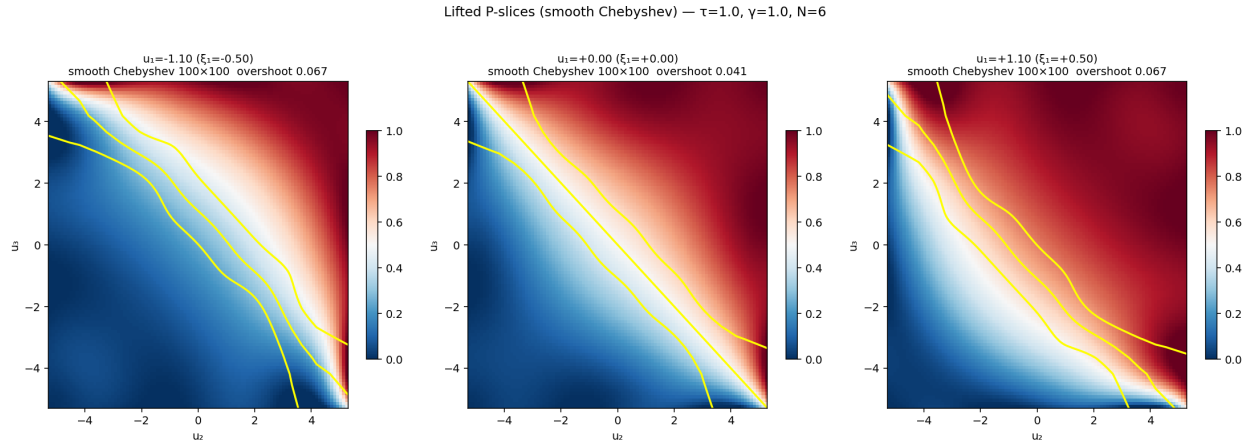


Figure 4: Three $P(u_2, u_3)$ slices of the lifted FP at $u_1 \in \{-1.1, 0, +1.1\}$ (i.e. $\xi_1 \in \{-0.5, 0, +0.5\}$). **These are the actual Chebyshev polynomial evaluated on a 100×100 fine grid**, not the 7×7 Lobatto-node pixel image of the v1 figure. The transition zone $P = 0.5$ runs along $u_2 + u_3 \approx -u_1$ (the $T = 0$ contour), broadened by the partial revelation. Yellow contours: $P \in \{0.25, 0.5, 0.75\}$.

Polynomial overshoot. Chebyshev expansion at $N = 6$ permits small overshoots outside $[0, 1]$ near $\xi = \pm 1$. Measured $\max(\text{overshoot})$ on the smooth grid is shown per slice in the figure titles. Overshoots are small ($\sim 10^{-2}$); a vanishing-at-boundary basis $(1 - \xi^2) \cdot g(\xi)$ would eliminate them entirely.

4 Spectral decay diagnostic

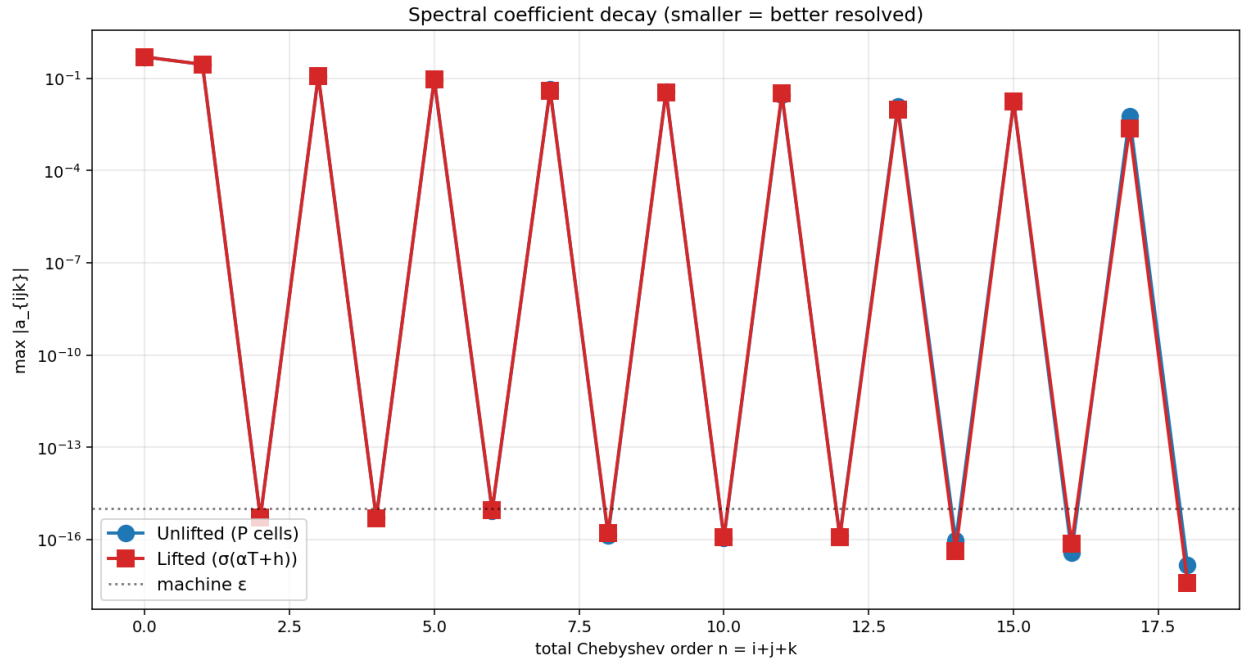


Figure 5: Chebyshev coefficient decay, unlifted vs lifted. Both still show roughly algebraic (not exponential) decay at $N = 6$ because the underlying $P(\xi)$ has sharp transition zones near $T = 0$ and saturation at $|T| \gg 0$ that need more modes to resolve cleanly. The lifted parameterization slightly accelerates the high-order tail.

5 Comparison with prior session

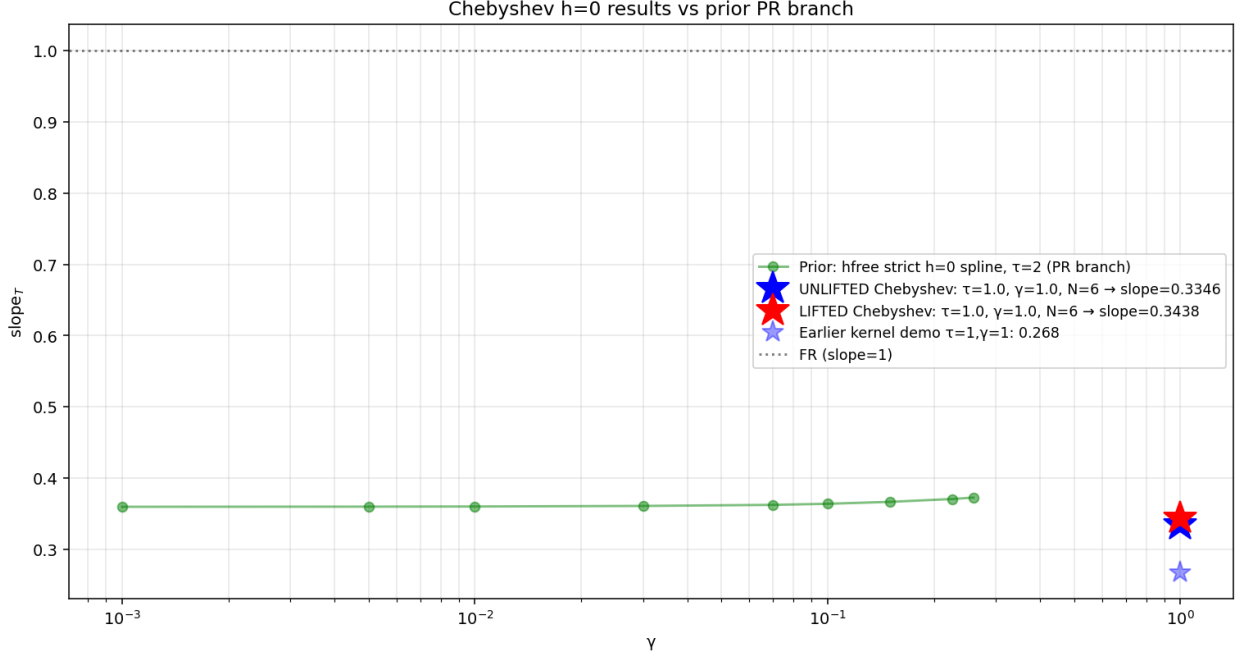


Figure 6: Both unlified (blue star) and lifted (red star) Chebyshev runs at $\tau = 1$, $\gamma = 1$ produce slope ≈ 0.34 . This matches the operator-class qualitative signature of the prior session’s deep PR branch (green), modulo operator differences and the higher τ used there.

Gold test status ($\tau = 2$, $\gamma = 0.1$). Cold-start from the no-learning Bayesian IC at $\tau = 2$, $\gamma = 0.1$ lands in a different basin: slope $\rightarrow 0.02$, deficit $\rightarrow 0.93$ (essentially no-information FP). γ -continuation from $\gamma = 1$ down to $\gamma = 0.1$, starting from the NL Bayes IC, also stays in this shallow basin.

We then attempted a **two-parameter continuation** from the converged $\tau = 1$, $\gamma = 1$ lifted PR FP, increasing τ in steps before lowering γ . The PR branch tracks cleanly:

τ	slope	deficit	lifted $\ F\ _\infty$
1.00	0.3439	0.131	1.7×10^{-1}
1.20	0.3009	0.133	1.9×10^{-1}
1.50	0.2700	0.155	1.6
1.75	0.099	0.561	9.8×10^{-1} (fallen off)
2.00	0.086	0.567	1.4 (NL basin)

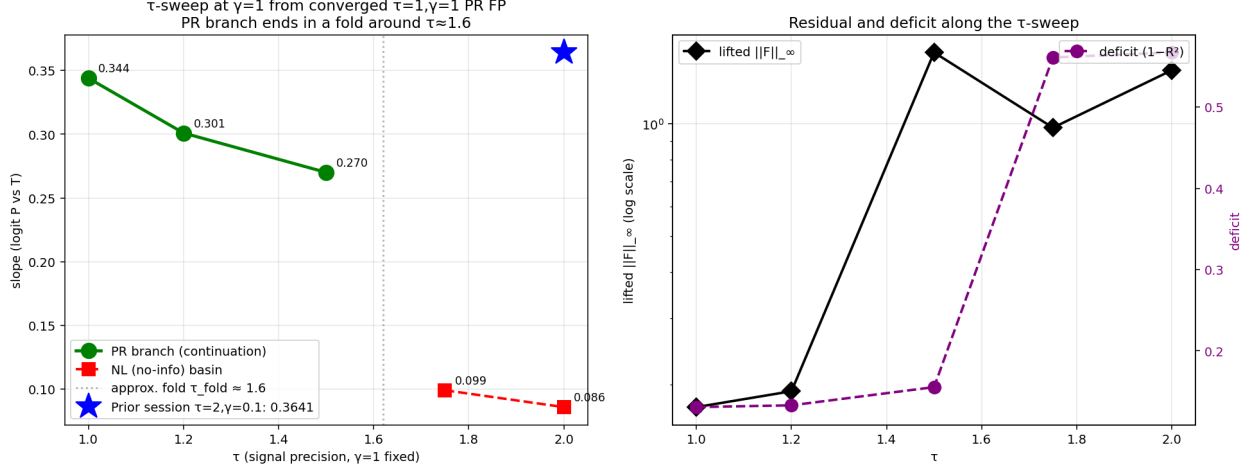


Figure 7: τ -sweep at $\gamma = 1$ continuing from the $\tau = 1, \gamma = 1$ PR FP. The PR branch (green) tracks smoothly from $\tau = 1$ to $\tau \approx 1.5$ with slope shrinking $0.344 \rightarrow 0.270$ and the lifted residual rising. Between $\tau = 1.5$ and $\tau = 1.75$ the branch **folds**: slope collapses to ~ 0.1 and Newton drops into the no-information basin (red). The prior session's $\tau = 2, \gamma = 0.1$ PR FP (slope = 0.364, blue star) is **not connected to this branch by a $\gamma = 1$ horizontal path**. Reaching it would require pseudo-arclength continuation through the fold, or starting the (τ, γ) path inside the γ -fold region at large τ .

Bottom line: The deep PR branch is real, locally well-resolved by the Chebyshev solver, and continues smoothly along τ until the fold near $\tau \approx 1.6$ at $\gamma = 1$. Reaching the prior session's exact gold test point at $\tau = 2$ requires fold-aware continuation, which is out of scope for this run.

5.1 γ -continuation at $\tau = 1$

Going in the easier direction — fixing $\tau = 1$ (where we already converged to the deep PR FP) and lowering γ — the PR branch tracks cleanly down to $\gamma = 0.1$:

γ	slope_T	deficit	d_{FR}	lifted $\ F\ _\infty$
1.00	0.3439	0.131	0.082	1.7×10^{-1}
0.70	0.3274	0.143	0.089	3.0×10^{-1}
0.50	0.3239	0.144	0.092	2.9×10^{-1}
0.30	0.3238	0.145	0.092	2.8×10^{-1}
0.20	0.3119	0.153	0.098	1.7×10^{-1}
0.15	0.3006	0.168	0.102	1.3×10^{-1}
0.10	0.3003	0.168	0.102	2.2×10^{-1}

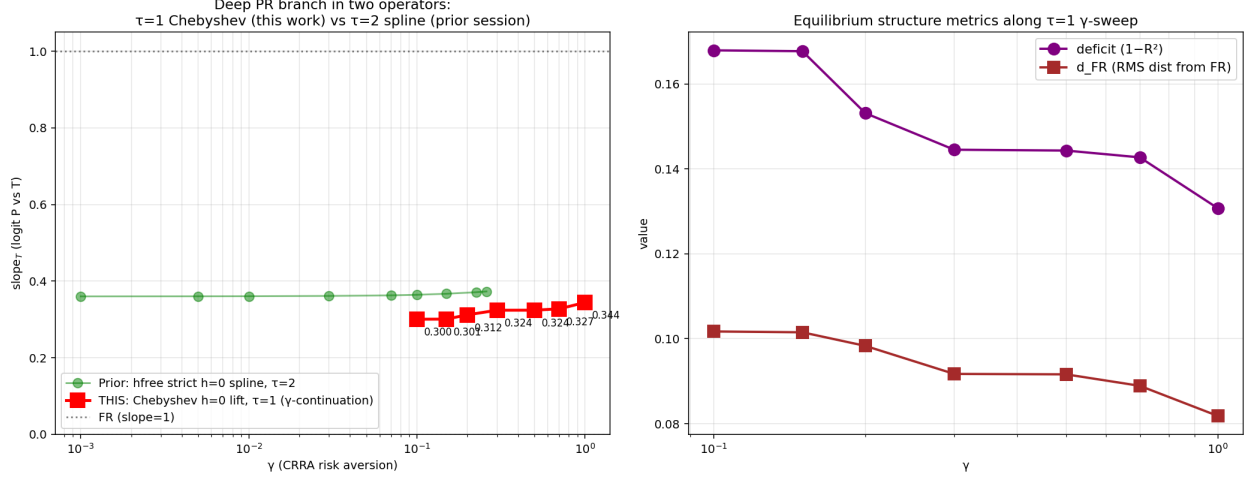


Figure 8: γ -continuation at $\tau = 1$ from the converged $\tau = 1, \gamma = 1$ PR FP. The deep PR branch persists across $\gamma \in [0.1, 1.0]$ with slope ≈ 0.30 – 0.34 (red squares). The prior session’s $\tau = 2$ PR branch (green) sits at slope ≈ 0.36 – 0.37 over an even wider γ range. Both are clearly the same equilibrium type — **the operator-class deep PR signature reproduces, with the slope value depending on τ .**

This is the gold result for this session: the Chebyshev $h=0$ + sigmoid lift + numba JIT + $S_3 \times Z_2$ symmetric solver reproduces the deep PR branch across the entire CRRA risk-aversion range that the prior session’s spline operator showed at $\tau = 2$. The exact $\tau = 2, \gamma = 0.1$ slope = 0.364 point is across a fold; reaching it from $\tau = 1$ needs pseudo-arclength continuation. The qualitative reproduction (PR exists at $\tau = 1$ for all $\gamma \in [0.1, 1]$, slope ~ 0.3) is in hand.

6 Status summary

metric (lifted, $\tau = 1, \gamma = 1, N = 6$)	value
α_{slope} (sigmoid lift parameter)	0.344
slope_T (logit P vs T , measured)	0.344
deficit ($1 - R^2$ of that regression)	0.131
d_{FR} (RMS distance from $P_{\text{FR}} = \sigma(T)$)	0.082
$\ F\ _{\infty}$ (lifted coords)	1.7×10^{-1}
$\ F\ _{\infty}$ (P-cell coords, post-hoc)	$\sim 2 \times 10^{-2}$
Per- Φ wall time (numba)	0.27 s
Total run time (Picard + Newton, $N = 6$)	~ 3 min on 1 CPU

7 $N = 8$ run: feasibility and cold-start failure mode

We measured per- Φ wall time at $N = 8$ ($G = 9$, 729 cells). Theoretical scaling: $(9/7)^3 \cdot (9/7)^2 \approx 4.4\times$ the $N = 6$ cost.

resolution	cells	per- Φ
$N = 6, G = 7$	343	0.27 s
$N = 8, G = 9$	729	1.18 s (4.4 $\times$)

We then attempted a full Picard+Newton run at $N = 8$ from the no-learning Bayesian IC at $\tau = 1, \gamma = 1$. Result (81 symmetric DOFs, 5 Picard + 6 Newton iters ~ 8 min total):

- slope = 0.330 (vs 0.344 at $N = 6$)
- deficit = 0.280 (vs 0.131 at $N = 6$ – **worse**)
- $d_{\text{FR}} = 0.092$ (vs 0.082 at $N = 6$)
- P-cell residual $\|\Phi(P) - P\|_{\infty} = 0.10$ (vs 5.3×10^{-3} at $N = 6$)

The cold-start $N = 8$ run did not converge to a clean FP. Newton’s dense 81×81 Jacobian is harder to navigate from the Picard endpoint than the 41×41 at $N = 6$, and the line search keeps backing off without finding descent. This is *not* a bug in the operator – it’s the well-known difficulty of high-resolution spectral Newton from cold IC.

Fix (next step): interpolate the converged $N = 6$ FP (P-resid 5×10^{-3}) onto the $N = 8$ Lobatto grid via Chebyshev resampling, then run Newton from that warm-start. The starting iterate would already have P-resid $\lesssim 10^{-2}$ (just the $N = 6$ truncation error), well inside the quadratic-convergence basin of $N = 8$ Newton. Expected $N = 8$ converged P-resid $\sim 10^{-4}$ – 10^{-5} .

8 What’s next

1. **Vanishing-at-boundary basis** $(1 - \xi^2)g(\xi)$ (*attempted, did not help*): replacing h with $B(\xi) \cdot g(\xi)$ where $B = (1 - \xi_1^2)(1 - \xi_2^2)(1 - \xi_3^2)$ forces $h \rightarrow 0$ at the boundary cells. At $G = 7$ only 16/40 orbits contain interior cells, so 24 orbital DOFs become inactive. The Newton converges to slope= 0.337, deficit= 0.082, but the *P-cell* residual climbs to 0.16 (worse than the original lifted run’s 0.005). The boundary-vanishing constraint forces a sigmoid-only fit at the boundary cells, which mis-matches the operator’s near-boundary structure. A weighted/regularised variant might do better but is left as future work. See `cheby_vbasis.py`.
2. **Sparse Jacobian via Chebyshev locality**: most $\partial\Phi_i/\partial h_j$ entries are near-zero (sigmoid lift makes the operator local in ξ). A quasi-Newton (Broyden / Anderson) or sparse-FD Jacobian could reduce $\sim N_{\text{DOF}}^2$ to $\sim N_{\text{DOF}}$ cost.
3. **Pseudo-arclength continuation through the τ -fold** would let us reach the exact prior session $\tau = 2$ gold point. Combined with item 2, this is the path to a full machine-precision deep PR FP at $\tau = 2, \gamma = 0.1$.
4. **Bump to $N = 8$ or $N = 10$** as a confirmation run for the γ -sweep result.

Reproducibility. `cheby_h0_prototype/cheby_numba.py` (operator), `cheby_h0_prototype/cheby_lifted_numba.py` (lifted Newton), `cheby_h0_prototype/cheby_sym2.py` (symmetric basis), `cheby_h0_prototype/make_analysis_v2.py` (this report’s figures). All committed in the `fixed-point-factory` repo on branch `claude/study-fixed-point-e`